Following the Information Bound from a Search-Window Prior with Sequential Bayesian Experiment Design in Nitrogen-Vacancy Magnetometry

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[Written last, once the simulation and lab results are in.]

I. INTRODUCTION

Nitrogen-vacancy (N-V) centers in diamond measure magnetic fields at room temperature with nanometer spatial resolution [1, 2], and are now used in settings ranging from condensed-matter physics to biology [3–5].

The most widely used readout is optically detected magnetic resonance (ODMR): the photoluminescence of the center is recorded while a microwave drive is stepped through a search window, the range of drive frequencies in which the spin resonance can lie, and the field is read from the position of the resonance. Every point of this scan takes time, so the number of measurements needed to locate the resonance to a given precision sets both the sensitivity and the bandwidth of the magnetometer. It matters most where the scan is repeated, as when a time-varying field is tracked [6] or a field is imaged with a spectrum recorded at every pixel [7, 8].

Locating the resonance is an instance of Hamiltonian learning: the parameters of a quantum system's Hamiltonian are inferred from measurements whose settings, here the microwave drive frequency, are chosen adaptively from what has been learned so far [9, 10]. For N-V centers, Hamiltonian learning and adaptive Bayesian estimation have been demonstrated mainly in the time domain, with Rabi and Ramsey sequences in which an evolution time or a readout phase is the chosen setting [11–14]. In ODMR, sequential Bayesian experiment design (SBED) has been applied directly [15]. SBED maintains a posterior distribution over the parameters, here represented by a sequential Monte Carlo (SMC) particle cloud, and chooses each drive frequency by maximizing an expected utility [9], here the expected information gain [15]; it reaches a given precision more than an order of magnitude faster than a sweep, the uniform scan conventionally used [15].

Non-adaptive strategies reduce the number of points instead, by measuring a subset of the window and reconstructing the spectrum from it, with a low-complexity reconstruction [16] or with compressed sensing, which matches the accuracy of a full scan with a small fraction of its points [6]. Learned measurement policies [17, 18] have recently spread with reinforcement learning (RL) across quantum metrology [19–23], including N-V magnetometry [24–26].

In ODMR, adaptive and non-adaptive strategies have been shown to need far fewer measurements than a sweep, and the gain is measured against that sweep [6, 15]; learned strategies are measured against the best established heuristics [17, 21, 26]. Information bounds are already used as benchmarks in Hamiltonian learning: the Bayesian Cramér–Rao bound certifies how close an SMC estimator comes to the best precision for the measurements it has chosen [9]. The same kind of bound, the van Trees inequality [27, 28], has been tested, for a regular prior, against the measured error of a photonic interferometer, which it captures when few photons are available [29]. In simulations of iterative single-molecule localization, where each measurement is placed using the previous estimate, it has been used to compare placement strategies [30]. Such a bound has not yet been applied to the ODMR search as a whole, starting from nothing more than the search window. It would give a strategy-independent reference, showing how close current strategies come to the best achievable performance and which of their measurements are unavoidable. It would also supply a mechanism for the speedup over a sweep, including its expected growth as the line occupies a smaller fraction of the window [15].

Here we derive this limit for the ODMR search and use it to analyze SBED and drive-frequency sweeps for continuous-wave (CW) and pulsed ODMR. The limit follows from Bayesian estimation theory with the search window as the prior [27, 28], and the information it counts is the same quantity that SBED maximizes. It covers the whole run: finding the resonance, narrowing in on it and refining the estimate are limits of a single bound rather than separate mechanisms. The same bound gives the speedup over a sweep: for a precise target it is approximately the ratio of the window width to the linewidth, independent of the signal-to-noise ratio, which explains its growth with the window. It also recovers the standard CW-ODMR sensitivity as the refine limit, reached only by a strategy that measures at the steepest slope of the line, and shows that the window limits the bandwidth while the line shape and the signal-to-noise ratio limit the sensitivity.

Section II derives the bound from the physical model, together with the speedup, sensitivity and bandwidth that follow from it. Section III describes SBED, with its particle filter, as a strategy that follows the bound, and the sweeps it is compared with, both evaluated against the bound. Section IV presents the simulations and the experiment, and Sec. V discusses the limits of the approach. The Supplemental Material [31] gives the ODMR line shapes, the derivations behind the bound for a search window and behind the expected information gain in each

regime, and the algorithm settings.

II. THEORY

A. Physical model

Under green illumination the negatively charged N-V center emits red photoluminescence (PL). A center in the ground-state spin sublevels $m_s = \pm 1$ crosses more often into a dark metastable singlet than one in $m_s = 0$, so it emits less PL, and the singlet decays preferentially back to $m_s = 0$ [3, 32]. Continuous illumination therefore pumps the spin into $m_s = 0$, and the spin state can be prepared and read out optically (Sec. S1 of the Supplemental Material [31]).

The ground state is a spin 1. With a static field B_z along the N-V axis, its Hamiltonian is [3]

$$H = D\,S_z^2 + \gamma_e B_z S_z + A_\parallel S_z I_z, \tag{1}$$

written in frequency units ($h = 1$), with $D \approx 2.87$ GHz the zero-field splitting, $\gamma_e \approx 28$ GHz/T the electron gyromagnetic ratio, and $A_\parallel$ the axial hyperfine coupling to the nitrogen nuclear spin I ($|A_\parallel| = 2.15$ MHz for ^{14}N, $I = 1$, and 3.03 MHz for ^{15}N, $I = 1/2$ [33]). The eigenstates are the product states $|m_s, m_I\rangle$, and the magnetic-dipole transitions $m_s = 0 \to \pm 1$, which conserve m_I, lie at

$$f_\pm(m_I) = D \pm \left(\delta_Z + A_\parallel m_I\right), \qquad \delta_Z = \gamma_e B_z, \tag{2}$$

where the Zeeman shift δ_Z is the distance of each transition from D: the field moves the two transitions apart, symmetrically about D. The hyperfine term splits each of them further into $2I+1$ lines, one per nuclear projection m_I, spaced by $|A_\parallel|$: a triplet for ^{14}N and a doublet for ^{15}N. Every N-V center contains one of the two isotopes (^{14}N at 99.6% natural abundance, ^{15}N in isotopically enriched or implanted samples), so the hyperfine structure is always present; whether it is resolved depends on the linewidth.

An ODMR measurement sets the microwave drive frequency f and records the PL, normalized to its off-resonance level. When f meets a transition, the drive transfers population from $m_s = 0$ to the darker $m_s = \pm 1$ state and the PL dips. When the two transitions, $2\delta_Z$ apart, are well separated compared with the linewidth, a drive near f_- couples only $m_s = 0$ and $m_s = -1$, and the spin reduces to a driven two-level system [34]. It is therefore enough to model one transition, and we take the one measured in our setup, f_- (Sec. III D); f_+ behaves the same way, mirrored about D. With the zero-field splitting D treated as a known constant, f_- carries the field, $D - f_- = \delta_Z = \gamma_e B_z$, and an uncertainty in f_- is the same uncertainty in the Zeeman shift.

In continuous-wave (CW) ODMR the laser and the microwave drive act on the spin at the same time. The laser pumps the spin back toward $m_s = 0$ at a rate Γ_p, and each optical cycle destroys the phase between $m_s = 0$ and $m_s = -1$, at a rate Γ_c; both rates grow with the laser power and saturate at high power. The spin's environment adds its own relaxation rate $1/T_1$ to the first and its dephasing rate $1/T_2$ to the second, so the populations relax at $\Gamma_1 = \Gamma_p + 1/T_1$ and the coherence at $\Gamma_2 = \Gamma_c + 1/T_2$. With these rates, and the angular Rabi frequency $\Omega_R = 2\pi\Omega$ of the drive, the steady-state line is a Lorentzian (Sec. S1 D of the Supplemental Material [31]), of half width [34]

$$\gamma = \frac{1}{2\pi}\sqrt{\Gamma_2^2 + \Omega_R^2\,\frac{\Gamma_2}{\Gamma_1}}. \tag{3}$$

At weak drive the laser sets the width, at strong drive the microwave. The spin bath also spreads the transition frequency itself, by $\sim 1/T_2^*$; this adds a Gaussian broadening to the Lorentzian, and it sets the width once both powers are low (Sec. S1 D of the Supplemental Material [31]). Whatever its origin, the line is one dip of a single effective width, taken here as a Lorentzian: the normalized PL expected at drive frequency f is

$$\mu(f) = 1 - C\,\frac{\gamma^2}{\gamma^2 + (f - f_-)^2}, \tag{4}$$

with contrast C, the depth of the dip at $f = f_-$, half width γ of Eq. (3) and full width at half maximum $\Gamma = 2\gamma$. Optical and microwave broadening are not modeled separately: they enter only through the single linewidth Γ, which is inferred with the line.

Equation (4) is a single line. It holds when the hyperfine lines merge, $\Gamma \gg |A_\parallel|$, or when the nuclear spin is polarized into one m_I [35].

In pulsed ODMR the laser is off while a single microwave π-pulse of duration $t_\pi = 1/(2\Omega)$ flips the spin, which removes the laser from the width of the line [34]. The dip then follows the Rabi response at the end of the pulse (Sec. S1 E of the Supplemental Material [31]),

$$\mu(f) = 1 - C\,\frac{\Omega^2}{\Omega_{\rm gen}^2}\,\sin^2\!\Big(\frac{\pi}{2}\,\frac{\Omega_{\rm gen}}{\Omega}\Big), \qquad \Omega_{\rm gen} = \sqrt{\Omega^2 + (f - f_-)^2}, \tag{5}$$

with $\Omega_{\rm gen}$ the generalized Rabi frequency. This line is not a Lorentzian: its width is the Fourier limit of the pulse, $\sim 1/t_\pi$, and it has side lobes up to 12% of the depth, dips where there is no resonance.

Noise. Each reading of the normalized PL is noisy. Its floor is photon shot noise: a reading that collects $N_{\rm ph}$ photons on average off resonance has a relative standard deviation $N_{\rm ph}^{-1/2}$, and slightly less inside the dip, where fewer photons arrive. Technical noise (laser intensity, microwave power, collection drift) adds to it [36]. We model a single reading, a shot, as

$$y = \mu(f) + \epsilon, \qquad \epsilon \sim \mathcal{N}(0, \sigma^2), \tag{6}$$

with one $\sigma \geq N_{\rm ph}^{-1/2}$ at every drive frequency; this neglects the small reduction of the shot noise inside the dip, of relative order of the contrast C. A measurement is the

mean of k shots at the same f, so its noise is $\sigma/\sqrt{k}$. The signal-to-noise ratio of a single shot depends on where it is taken: it is the drop in PL at f over the noise,

$$\mathrm{SNR}(f) = \frac{1-\mu(f)}{\sigma} = \frac{C}{\sigma}\,\frac{\gamma^2}{\gamma^2+(f-f_-)^2}. \tag{7}$$

It is largest on resonance, where the drop is the contrast C,

$$\mathrm{SNR} \equiv \mathrm{SNR}(f_-) = C/\sigma, \tag{8}$$

the contrast over the noise, the single number by which a line is labeled. All SNR values in this paper follow Eq. (8).

The prior knowledge of the experimentalist is an interval of width W that holds the resonance, $f_- \in [f_{\min}, f_{\min}+W]$, and the drive frequency is scanned over at least this interval. The ratio W/Γ, the inverse of the fraction of the window the line occupies, sets how hard the search is.

The model above predicts what a measurement records for given parameters. The rest of this section turns the question around: given the measurements, how well can f_- be known, how much can each measurement add, and where should the next one be taken.

B. Information of a measurement

We use the notation of experiment design [37]: the design $d \equiv f$ is the drive frequency at which a measurement is taken, the outcome y is the normalized PL it records, the parameter of interest $\theta \equiv f_-$ is the transition frequency, and $\phi = (\Gamma, C)$ are the nuisance parameters. The expected normalized PL $\mu(d;\theta,\phi)$ of Eq. (4), now written as a function of the design and the parameters, is the mean outcome, and Eq. (6) gives the spread about it.

A measurement at drive frequency d returns one number y, the normalized PL averaged over its k shots. The model predicts the signal $\mu(d;\theta,\phi)$, and by Eq. (6) y scatters about it with variance σ^2/k, the same whatever θ is. The likelihood $\mathcal{L}(\theta) = p(y \mid d,\theta)$ is the probability of that outcome if the dip is at θ. For the Gaussian scatter of Eq. (6),

$$\mathcal{L}(\theta) = \sqrt{\frac{k}{2\pi\sigma^2}}\,\exp\Big[-\frac{k\,\big(y-\mu(d;\theta,\phi)\big)^2}{2\sigma^2}\Big].$$

How much one measurement at d tells about θ is its Fisher information. With the score $\partial \ln\mathcal{L}/\partial\theta = (k/\sigma^2)(y-\mu)\,\partial\mu/\partial\theta$ it is

$$\begin{aligned} I(d) &= \Big\langle \Big(\frac{\partial \ln\mathcal{L}}{\partial\theta}\Big)^2\Big\rangle_y = \Big(\frac{k}{\sigma^2}\,\frac{\partial\mu}{\partial\theta}\Big)^2 \underbrace{\langle (y-\mu)^2\rangle_y}_{\sigma^2/k} \\ &= \frac{\overbrace{k}^{\text{shots}}}{\underbrace{\sigma^2}_{\text{shot variance}}}\;\overbrace{\Big(\frac{\partial\mu(d;\theta,\phi)}{\partial\theta}\Big)^2}^{\text{slope}^2}, \end{aligned} \tag{9}$$

the average running over the outcomes y the measurement could return, and the slope $\partial\mu/\partial\theta$ being how fast the predicted signal moves when θ moves.

For the Lorentzian of Eq. (4) the slope vanishes on resonance, where the dip is flat, and on the baseline far from it: a measurement at the bottom of the dip or far outside it says almost nothing about where the dip is. The slope is steepest on the two sides of the dip, a detuning $\gamma/\sqrt{3}$ from the resonance, where the dip is 3/4 deep (Appendix A). A measurement there carries the most information one measurement can,

$$I_{\max} = \max_d I(d) = \frac{27}{16}\,k\,\frac{\mathrm{SNR}^2}{\Gamma^2}, \tag{10}$$

with $\mathrm{SNR} = C/\sigma$ of Eq. (8). The best place to measure is thus the flank of the dip, not its center, and the information there grows as $(\mathrm{SNR}/\Gamma)^2$: a deeper or a narrower line is equally valuable. $I_{\max}$ assumes the nuisance parameters (width, contrast) are known; with free nuisances it is an upper bound on the usable information. The nuisances are inferred along with θ but are not the subject of this work.

Measurements are independent given θ, so their Fisher information adds: n measurements at $d_1,\ldots,d_n$ carry

$$J_n = \sum_{i=1}^{n} I(d_i) \;\le\; n\,I_{\max},$$

with equality only when every measurement is taken at the steepest slope of the true line. By the Cramér–Rao bound [38], an unbiased estimate of θ from them has a standard deviation of at least $J_n^{-1/2} \ge (nI_{\max})^{-1/2}$.

C. Prior information

Fisher information measures knowledge in units of an inverse variance of θ, which is Hz^{-2} since θ is a frequency: a Gaussian estimate of standard deviation σ_θ is worth $J = \sigma_\theta^{-2}$. The same currency measures what is known *before* the first measurement, and that depends on the task.

In a *global search* the resonance is known only to lie somewhere in the window of width W, and the state of knowledge is a uniform prior $p(\theta) = 1/W$ on the window. Its spread is its standard deviation σ_0; with v the distance from the center of the window,

$$\sigma_0^2 = \int_{-W/2}^{W/2} \frac{v^2}{W}\,dv = \frac{W^2}{12}. \tag{11}$$

In *tracking* the resonance was measured a short time ago and has since moved by a random amount, so the prior is a Gaussian of standard deviation σ_0 around the last estimate, narrow compared with the window.

The Fisher information of the prior probability density function (PDF) $p(\theta)$ is $J(p) = \int p'(\theta)^2/p(\theta)\,d\theta$.

For the Gaussian prior of tracking, $J(p) = \sigma_0^{-2}$. The uniform prior of a global search steps from $1/W$ to zero at the edges of the window, where p' diverges, so $J(p) = \int p'(\theta)^2/p(\theta)\,d\theta = \infty$ and the bound of Eq. (13) becomes empty. We use instead, among all densities confined to a width W, the one with the least information, $p(v) = (2/W)\cos^2(\pi v/W)$, with $J(p) = 4\pi^2/W^2$ [28] (Sec. S2 of the Supplemental Material [31]). The information that enters the bound is therefore

$$J_0 = \begin{cases} 4\pi^2/W^2 & \text{global search,} \\ \sigma_0^{-2} & \text{tracking.} \end{cases} \tag{12}$$

For a global search the $\cos^2$ density is only a device for the bound. The prior the strategy uses stays uniform, with spread $\sigma_0 = W/\sqrt{12}$, and the bound then limits the worst-case error over the window rather than the error averaged over the uniform prior. In both cases J_0 is knowledge the strategy starts with: a narrower prior holds more of it, and no measurement had to pay for it.

D. Van Trees bound

What we compare the strategies against is a limit on the uncertainty, whatever its source, the prior or the measurements. The van Trees inequality [27, 28] adds the information of the two, and with $I(d) \le I_{\max}$ gives

$$\sigma_\theta(n) \;\ge\; \sigma_{\mathrm{vT}}(n) = \left(J_0 + nI_{\max}\right)^{-1/2}, \tag{13}$$

for any estimator, biased or not. Here $\sigma_\theta^2(n) = \mathbb{E}\big[(\hat\theta_n - \theta)^2\big]$ is the mean squared error (MSE) of the estimate $\hat\theta_n$ after n measurements, and σ_θ its root. It is an error against the true θ, not the uncertainty a strategy claims; what it is averaged over depends on the task, as discussed below.

The curve has two limits, set by how the information of the measurements compares with that of the prior. While $nI_{\max} \ll J_0$, it stays at the prior, $\sigma_{\mathrm{vT}} \approx J_0^{-1/2}$: the measurements have not yet added to what the prior told. An estimate in this limit leans on the prior and is biased toward its center, which is why its error may lie below the unbiased Cramér–Rao bound $(nI_{\max})^{-1/2}$. Once $nI_{\max} \gg J_0$, the prior is washed out and $\sigma_{\mathrm{vT}} \approx (nI_{\max})^{-1/2}$, the unbiased Cramér–Rao bound of Sec. II B, which falls with slope $-\frac{1}{2}$ on a log–log scale and is reached only by measuring every time at the steepest slope. The two limits cross at $n_c = J_0/I_{\max}$.

Any strategy therefore starts in the prior limit and, if it does well, ends in the data limit, and in between it must pass from one to the other. We name these regimes after what the strategy does in them: *search* while it has yet to find the dip, so that its measurements carry little information and its error stays near the prior; *refine* once it measures on the sides of a dip it has located, so that they carry about $I_{\max}$ and its error follows the data limit; and *localize* in between, from the first detection of the dip until it locks onto the dip, as it narrows down from the window to the dip. After that a global search has become a tracking problem: its posterior is roughly Gaussian of width $\lesssim \Gamma$ around the dip, and it behaves like a tracking run. A tracking run with $\sigma_0 \lesssim \Gamma$ has the dip located from the start, so it has no search and is all refine.

For a wide window, a global search reaches the data limit far later than the information per measurement alone would suggest. If every measurement carried $I_{\max}$, the crossover would come after

$$n_c = \frac{J_0}{I_{\max}} = \frac{64\pi^2}{27}\,\frac{(\Gamma/W)^2}{k\,\mathrm{SNR}^2} \overset{W\gg\Gamma}{\ll} 1 \tag{14}$$

measurements. But $I_{\max}$ is reached only once the dip has been found: a measurement on the baseline carries $I(d) \approx 0$, so the information collected stays near that of the prior while the strategy searches. What such a measurement contributes instead is *elimination*: it shrinks the prior itself.

Elimination model. To see the three regimes in an actual error curve, consider the simplest strategy that has all of them. It steps through the window at spacing Δd (e.g. $\Delta d = \Gamma/3$) only until it finds the dip, and from then on measures only at the steepest slope. This is not a sweep, which keeps spreading its measurements over the whole window after the dip is found (Sec. II G). Let the SNR be so high that a single measurement taken within the line, $|d-\theta| \lesssim \gamma$, shows a drop that cannot be mistaken for noise, so the first measurement that lands in the dip detects it. After n measurements without a dip, the resonance is known only to lie in the part of the window not yet excluded, and the posterior is uniform there, so

$$\sigma_\theta(n) = \frac{W - n\Delta d}{\sqrt{12}}, \qquad n < n_d, \tag{15}$$

where $n_d \le W/\Delta d$ is the step that finds the dip. Being a model of the actual error rather than a bound, it uses the standard deviation of the part of the window not yet excluded, Eq. (11) with W replaced by $W - n\Delta d$, and not the J_0 of Eq. (12). In inverse variance, $\sigma_\theta(n)^{-2} = 12/(W - n\Delta d)^2$: the gain per step is not constant but grows as the remaining width shrinks. Detection localizes the resonance to about one grid step, a uniform interval of width Δd and inverse variance $12/\Delta d^2$, after which each measurement adds $I_{\max}$:

$$\sigma_\theta(n) = \left(\frac{12}{\Delta d^2} + (n - n_d)\,I_{\max}\right)^{-1/2}, \qquad n \ge n_d. \tag{16}$$

Eqs. (15)–(16) start at the true error $\sigma_\theta = \sigma_0$ at $n = 0$, above the bound, and reach its data limit $\sigma_\theta \to (nI_{\max})^{-1/2}$ for $n \gg n_d$. In between, each regime takes a simple shape (Fig. 1, solid curve): search, Eq. (15), is a plateau, since a linear fall looks flat on a log–log axis until about half the window is excluded; localize is the single detection step, which drops σ_θ from about the window

to about Δd; refine, Eq. (16), is the $-\frac{1}{2}$ tail. In a window about as narrow as the line the first measurements already land in the dip, so there is no plateau (dashed curve). A sweep has none of the three, since it gains the same information I_{sweep} at every step (Sec. II G).

Reading the bound. We judge a strategy by how close its error, averaged over runs, comes to Eq. (13): a strategy on the curve gets all the information its measurements can give, and the gap above it shows how much it loses. How a gap is read depends on the task, because the error the bound limits is not always the error the runs report. In tracking the two are the same: both are averaged over the Gaussian prior, and the bound starts at the true error σ_0. An averaged error above it means that the measurements carried less than $I_{\max}$, or that the strategy did not use them efficiently.

In a global search the bound limits the error of the worst-placed resonance in the window, while our runs report the error averaged over a uniformly placed one. Before any measurement the average error, $W/\sqrt{12}$, lies $\pi/\sqrt{3} \approx 1.8$ times above the bound; this gap belongs to the window and not to any strategy. It closes around n_c: the best achievable average error is within 10% of the bound from $n \approx 2\,n_c$, and from $n \approx 10\,n_c$ it lies below it by at most about 1% (Sec. S2 of the Supplemental Material [31]).

A strategy is therefore read against the curve through its error averaged over runs, not through a single run. In one run the dip is found wherever it happens to lie, possibly at the first measurement, so where that run's error falls relative to the bound says little about the strategy. Averaged over runs, the gap above the curve has two parts. Before the dip is found, beyond the part that belongs to the window, it measures how fast the strategy searches. After the dip is found, the slope of the averaged curve gives the information each measurement carries, a fraction $q \leq 1$ of $I_{\max}$ (Sec. II G), and a gap that does not close means $q < 1$. A claimed uncertainty below the curve by more than about 1% plus the scatter of the average points to an overconfident posterior.

The bound holds for the MSE whatever the shape of the posterior, and for the posterior mean that error equals the expected posterior variance. The posterior standard deviation, averaged over runs, can therefore be compared with σ_{vT} from the first measurement on, as long as the posterior is calibrated.

The bound treats the width Γ and the contrast C as known. Inferring them as well can only add to the uncertainty in θ, but it adds nothing when the measurements are placed symmetrically on the two sides of the dip: a shift of θ then raises the signal on one side and lowers it on the other, while a change of Γ or C changes both sides alike, so the data tell the two apart.

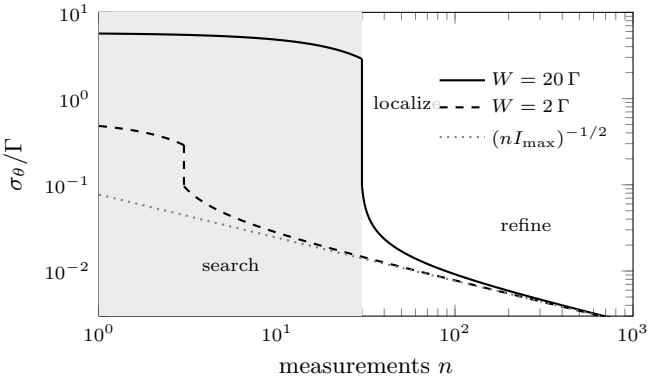


FIG. 1. Schematic of the three regimes of an adaptive strategy, drawn with the elimination model, Eqs. (15)–(16); a Bayesian adaptive strategy follows the same profile, with a less sharp localize step. Illustrative values (SNR 10, $k = 1$, $\Delta d = \Gamma/3$, dip found halfway through the grid). Solid: wide window, $W = 20\,\Gamma$; dashed: narrow window, $W = 2\,\Gamma$; dotted: unbiased Cramér–Rao bound $(nI_{\max})^{-1/2}$. The wide window shows a search plateau (shaded), a localize cliff at detection (vertical segment) and the $-\frac{1}{2}$ refine tail; the narrow window shows no plateau. Both start near the prior width σ_0 and tend to the data limit of the van Trees bound, the unbiased Cramér–Rao bound, at large n. Simulated and measured runs are shown against the bound in Sec. IV.

E. Bayesian adaptive strategy

A Bayesian strategy such as SBED is defined quite differently from the elimination model: it keeps a posterior over θ and picks each drive frequency by the expected information gain (Sec. II F), with no grid and no detection rule. Its error nevertheless follows the same three-part profile. Before the dip is found, a measurement on the baseline only removes the posterior mass near where it was taken, so the posterior shrinks by elimination, as in the model; once a measurement lands in the dip the posterior collapses onto it; and once the dip is located, the strategy measures at the steepest slope and reaches the data limit (Sec. II F). The model is therefore a picture of a Bayesian run, not a prediction of its step counts: at low SNR one measurement no longer reveals the dip, n_d grows, and localize spreads over many measurements. Eq. (13) holds at any SNR and needs no detection model, so the runs are read against it alone.

F. Information gained per measurement

The bound counts information as Fisher information, an inverse variance of θ. A Bayesian strategy chooses its measurements by a different quantity, how much a measurement is expected to change the posterior. This section defines that quantity, the form of it that SBED computes, and when maximizing it reaches the bound.

Expected information gain. How much a distribution q differs from a distribution p is measured by the Kullback–Leibler divergence [39], $D_{\mathrm{KL}}(q\,\|\,p) = \int q \ln(q/p)$, which is zero only when the two agree. What a measurement at d contributes is the divergence of the posterior after it from the posterior before it. Before the outcome is known, its expectation over the outcomes y the measurement could return is the expected information gain [40],

$$\begin{aligned}\mathrm{EIG}(d) &= \mathbb{E}_{y|d}\Big[D_{\mathrm{KL}}\big(p(\theta,\phi\mid y,d)\,\|\,p(\theta,\phi)\big)\Big]\\ &= I(\theta,\phi;y\mid d)\\ &= \underbrace{H[y\mid d]}_{\text{predictive entropy}} - \underbrace{\mathbb{E}_{\theta,\phi}H[y\mid\theta,\phi,d]}_{\frac{1}{2}\ln(2\pi e\,\sigma^2/k)\ \text{at every } d}\ , \end{aligned}\tag{17}$$

the mutual information [41] between the parameters and the outcome, with $p(\theta,\phi)$ the current posterior and H the entropy. The second term is the entropy of the noise of Eq. (6), so all the dependence on d is in the predictive entropy.

The variance form computed by SBED. The predictive distribution of y has two spreads, the noise σ^2/k and the spread of the predicted signal over the posterior, $\sigma^2_{\mathrm{pred}}(d) = \mathrm{Var}_{\theta,\phi}[\mu(d;\theta,\phi)]$. Its exact entropy needs the whole distribution. SBED replaces it with a Gaussian of the same variance, which has the largest entropy for a given variance [41], so the result is an upper bound on the gain,

$$\mathrm{EIG}(d) \le \tfrac{1}{2}\ln\Big(1+\frac{k\,\sigma^2_{\mathrm{pred}}(d)}{\sigma^2}\Big),\tag{18}$$

exact when the predicted signal is Gaussian distributed over the posterior. SBED (Sec. III B) evaluates Eq. (18) at every candidate drive frequency and measures where it is largest, the greedy rule

$$d_{n+1} = \arg\max_d \mathrm{EIG}_n(d),\tag{19}$$

with EIG_n evaluated on the posterior after n measurements.

Two questions. (A) Eq. (18) replaces the predictive distribution with a Gaussian: when does it agree with the exact gain? (B) The bound is reached only by measuring at the steepest slope, yet the greedy rule of Eq. (19) maximizes the expected gain and never refers to the slope: when does it put the measurements there? In refine the answer to both is "always"; in search and localize both fail.

Refine: the gain becomes Fisher information. Once the dip is located, the posterior of θ is a single peak, narrow compared with the line, so a first-order Taylor expansion about the estimate $\hat\theta$ makes the signal linear in θ: $\mu(d;\theta) \approx \mu(d;\hat\theta) + (\partial\mu/\partial\theta)(\theta-\hat\theta)$. Then $\sigma_{\mathrm{pred}}(d) = |\partial\mu/\partial\theta|\,\sigma_\theta$ whatever the shape of the peak, and the rule measures where the Fisher information $I(d)$ of Eq. (9) is largest, at the steepest slope. With enough data the peak also becomes Gaussian by the central limit theorem for posteriors (the Bernstein–von Mises theorem [42]), of variance $\sigma_\theta^2 \approx 1/J_n$, so Eq. (18) is an equality and

$$\begin{aligned}\mathrm{EIG}(d) &= \tfrac{1}{2}\ln\Big(1+\underbrace{\frac{k}{\sigma^2}\Big(\frac{\partial\mu}{\partial\theta}\Big)^2}_{I(d)}\ \underbrace{\sigma_\theta^2}_{1/J_n}\Big)\\ &= \tfrac{1}{2}\ln\frac{\overbrace{J_n + I(d)}^{J_{n+1}}}{J_n} = \ln\frac{\sigma_{\theta,n}}{\sigma_{\theta,n+1}}\ :\end{aligned}\tag{20}$$

the measurement adds $I(d)$ to the information, and the EIG is the logarithm of the shrinkage of the posterior width. The variance form and the exact gain then pick the same point, which answers (A). Each gain is a mutual information, Eq. (17), so by the chain rule of mutual information [41] the expected gains of a run add up to the information all its outcomes carry together. In refine each depends on its measurement only through $I(d_i)$, so maximizing every $I(d_i)$ also maximizes the total: greedy EIG is optimal over the whole run and reaches the data limit of Eq. (13), which answers (B) (Sec. S3 of the Supplemental Material [31]). Both answers need the posterior to be one peak with $\sigma_\theta \ll \gamma$ and almost no mass left elsewhere in the window, which holds in refine and not at the start of localize (Sec. S4 of the Supplemental Material [31]).

Search and localize. Before the dip is located the posterior of θ is spread over the window, the predicted signal at d is two-valued (the dip is there or not), and both answers fail (Sec. S3 of the Supplemental Material [31]). For (A), the variance form ranks search candidates as the exact gain does but overvalues them against refine candidates, and in localize it keeps spending measurements on residual posterior mass far from the dip, which lowers the MSE rather than the entropy. For (B), where one measures now changes what later measurements can gain, so choosing the most informative next measurement need not give the most informative run. The bound is therefore met in refine, and the strategies of Sec. III B differ in how they pass search and localize.

G. Speedup over a drive-frequency sweep

The same budget predicts the speedup over a conventional sweep. An adaptive strategy in the refine regime measures at the steepest slope and gains, per measurement, Eq. (10),

$$I_{\max} = \frac{27}{16}\,\frac{k\,C^2}{\sigma^2\,\Gamma^2}.$$

A sweep spreads its measurements uniformly over the window, so the information it gains per step is the window average of Eq. (9), taken over the drive frequency in

the window (not over the prior),

$$I_{\text{sweep}} = \frac{1}{W}\int I(f)\,df = \frac{k}{\sigma^2 W}\int \left(\frac{\partial\mu}{\partial\theta}\right)^2 df \overset{W\gg\Gamma}{=} \frac{\pi}{2}\,\frac{k\,C^2}{\sigma^2\,\Gamma\,W}. \tag{21}$$

Comparing the two rates,

$$I_{\max} \propto \frac{k\,C^2}{\sigma^2}\,\frac{1}{\Gamma^2}, \qquad I_{\text{sweep}} \propto \frac{k\,C^2}{\sigma^2}\,\frac{1}{\Gamma}\,\frac{1}{W}:$$

they share the factor kC^2/σ^2 and differ only in the width factor. The extra factor Γ/W of the sweep is the fraction of its measurements that land on the line at all.

Where a sweep measures does not depend on what it has learned, so its information grows at the constant rate I_{sweep}, while an adaptive strategy in refine gains $I_{\max}$ per measurement. For a precise target, where the search cost no longer matters, the speedup is the ratio of the two rates,

$$R_\infty = \frac{I_{\max}}{I_{\text{sweep}}} = \frac{27}{8\pi}\,\frac{W}{\Gamma} \approx 1.07\,\frac{W}{\Gamma}: \tag{22}$$

the speedup is the inverse occupancy of the window, the number of linewidths that fit in it, which Dushenko *et al.* [15] identified qualitatively; Eq. (22) gives its prefactor. It is independent of the SNR, because both strategies gain information $\propto \text{SNR}^2$. For a loose target the search cost dominates and the speedup is smaller (Sec. S5 of the Supplemental Material [31]). A sweep cannot approach the van Trees bound at any n: its rate I_{sweep} is fixed by the window, while the bound's data limit requires every measurement at the steepest slope.

Comparing strategies. A strategy is thus described by two numbers: what it pays to find the dip, and the rate at which it gains information afterwards. Both are read off the error averaged over runs, without a detection model; a single run's values depend on where its dip happened to lie and prove nothing on their own. Late in a run the inverse of the averaged squared error grows linearly,

$$\sigma_\theta(n)^{-2} \approx q\,I_{\max}(n - n_0), \tag{23}$$

where $q \le 1$ is the fraction of $I_{\max}$ that the measurements carry, and n_0 is the number of measurements lost, on average, before this growth began. The offset n_0 generalizes n_d, the step that finds the dip in the elimination model of Eqs. (15)–(16): it holds the search and also whatever localizing costs beyond the single step of the elimination model. A sweep has $q = I_{\text{sweep}}/I_{\max} = 1/R_\infty$ and $n_0 \approx 0$, and the ideal adaptive strategy has $q = 1$ and $n_0 \approx n_d$, so all strategies can be placed on one (n_0, q) plane. The number of measurements to a precise target is then $n_0 + \sigma_{\text{tgt}}^{-2}/(qI_{\max})$. In the inverse squared error each strategy is a straight line that leaves the axis at n_0 with slope q (Fig. 2): the sweep starts at once but climbs slowly, an adaptive strategy starts late but climbs fast, so it wins for a precise target and can lose for a loose one.

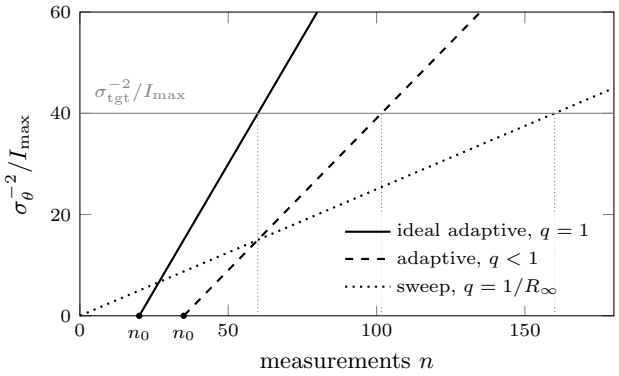


FIG. 2. Schematic of Eq. (23), with exaggerated values chosen for legibility (a real sweep has $q = 1/R_\infty \approx \Gamma/W$, far smaller). In the inverse squared error, in units of $I_{\max}$, each strategy is a line that leaves the axis at its offset n_0 with slope q. Solid: ideal adaptive strategy, $q = 1$, $n_0 = n_d$; dashed: an adaptive strategy that loses information, $q < 1$, larger n_0; dotted: sweep, $n_0 \approx 0$, $q = 1/R_\infty$. The vertical lines mark the measurements each needs to reach a precise target (horizontal line), $n_0 + \sigma_{\text{tgt}}^{-2}/(qI_{\max})$. For a loose target, below the crossings, the sweep is ahead.

H. Sensitivity and bandwidth

Sensitivity and bandwidth follow from the same budget once measurements are converted to time. Let one measurement take $t_m = k\,t_s$ (k shots of duration t_s), so that n measurements take $T = nt_m$. With D known, $D - f_- = \delta_Z = \gamma_e B_z$, so the field uncertainty is $\sigma_B = \sigma_{f_-}/\gamma_e$.

Sensitivity (refine). In the refine regime every measurement adds $I_{\max}$, so after a time T

$$\sigma_{f_-}(T) = (nI_{\max})^{-1/2} = \left(\frac{T}{t_m}\,I_{\max}\right)^{-1/2},$$

$$\sigma_{f_-}\sqrt{T} = \sqrt{\frac{t_m}{I_{\max}}} = \gamma_e\,\eta.$$

The uncertainty falls as $\sigma_{f_-} \propto T^{-1/2}$, so the product $\sigma_{f_-}\sqrt{T}$ does not depend on T and characterizes the strategy rather than the length of the run. Converted to field, this constant is the sensitivity η,

$$\eta = \sigma_B\sqrt{T} = \frac{1}{\gamma_e}\sqrt{\frac{t_m}{I_{\max}}} = \frac{1}{\gamma_e}\sqrt{\frac{16\,k\,t_s\,\Gamma^2}{27\,k\,\text{SNR}^2}} = \frac{4}{3\sqrt{3}}\,\frac{\Gamma\sqrt{t_s}}{\gamma_e\,\text{SNR}}, \tag{24}$$

with $\text{SNR} = C/\sigma$ the single-shot SNR of Eq. (8). The number of shots k cancels as well: averaging k shots into one measurement costs k times the time and gives k times the information. This is the standard CW-ODMR sensitivity of a Lorentzian line [34], here the refine limit of the budget, reached only by a strategy that measures at the steepest slope. A strategy whose refine measurements are

not all at the steepest slope gains less per measurement: if they carry on average a fraction $q \leq 1$ of $I_{\max}$, $I_{\max}$ is replaced by $qI_{\max}$ in Eq. (24), and its sensitivity is worse by $1/\sqrt{q}$, i.e. $\eta/\sqrt{q}$.

Sensitivity over a run. Read off a run of length T, the product $\sigma_B(T)\sqrt{T}$ is not yet this constant: during the search the error barely falls and the product grows as $\sqrt{T}$, and after detection it approaches η from above, with an excess that falls as $1/T$ (Sec. S5 of the Supplemental Material [31]).

Bandwidth (whole run). A field reading is useful once its error reaches the target σ_{tgt}; the time needed, $T(\sigma_{\mathrm{tgt}}) = t_m\,[n_0 + \sigma_{\mathrm{tgt}}^{-2}/(qI_{\max})]$ by Eq. (23), sets the bandwidth $1/T(\sigma_{\mathrm{tgt}})$. For a tight target the refine term dominates and gives the sensitivity of Eq. (24) again. For a loose target the search term dominates, $T \approx n_0 t_m$: the bandwidth cannot exceed about $1/(n_0 t_m)$, whatever the SNR (Sec. S5 of the Supplemental Material [31]). The window, i.e. the prior, therefore limits the bandwidth, while the line shape and SNR limit the sensitivity. Only the total integration time $T = nk\,t_s$ enters the bound, not how it is divided into measurements: n measurements of k shots carry the same information as nk measurements of one shot at the same drive frequency. Neglecting the overhead of switching the drive frequency, the cost of a run is therefore its total number of shots nk, the integration time in units of t_s, and strategies that group their shots differently are compared on that axis.

All closed-form results of this section are for the Lorentzian line of Eq. (4). For pulsed ODMR the same quantities follow, numerically, with Eq. (5) in place of Eq. (4). [Which ODMR mode the results focus on, once the simulations and the experiment are complete; where pulsed goes (a Theory subsection for the information of the pulsed line, and its own Results).]

III. METHODS

A. Bayesian inference

Every strategy runs the same loop: it chooses a drive frequency d_n, takes one measurement y_n there, updates the posterior over the line parameters with it, and stops when the stop rule of Sec. III B fires. The strategies differ only in the first step; the posterior and its update, described here, are shared. The unknowns are those of Sec. II B, the parameter of interest θ and the nuisances ϕ, together with the noise level σ, which is treated separately below.

The posterior over (θ, ϕ) is a weighted cloud of particles $\{(\theta_i, \phi_i), w_i\}_{i=1}^{N_p}$, updated by sequential Monte Carlo [43]. Each particle is one hypothesis for the line. It holds the Zeeman shift $\delta_{Z,i}$, so $\theta_i = D - \delta_{Z,i}$ with D fixed (the simulations measure the mirror transition $D + \delta_{Z,i}$, which is equivalent by symmetry), and the nuisances: the linewidth itself, as the half width $\gamma = \Gamma/2$ of Eq. (4), and the contrast C. The mechanisms that set the width, Eq. (3), are not inferred; only the width they produce is. For the pseudo-Voigt line the width is held as its Lorentzian half width and the standard deviation of its Gaussian part.

The noise is Gaussian, Eq. (6), as it is for the photon numbers of an ODMR shot, but its level σ is not known in advance: technical noise adds to the shot noise by an amount that depends on the setup and can drift. σ is therefore inferred with the line, but not as a particle dimension. Given a particle's line, the Gaussian likelihood is conjugate to an inverse-gamma prior on σ^2, so each particle carries that posterior in closed form, updated from its own residuals, and σ^2 is integrated out exactly: a Rao–Blackwellized particle filter with per-particle sufficient statistics [44, 45]. A measurement then reweights each particle by a Student-t predictive, the Gaussian likelihood averaged over that particle's noise posterior. Integrating out matters: a likelihood with a plug-in estimate of σ is largest for a particle whose σ happens to equal one residual, so resampling would favor such particles and drive σ to its lower bound, while the Student-t tails do not reward that. Old residuals are discounted, as in the variance discounting of dynamic linear models [46], so the noise posterior can follow a slow drift of σ.

When the weight has concentrated on a few particles, the cloud is resampled systematically [47] and jittered with the Liu–West kernel [48], which shrinks every particle towards the mean of the cloud and adds Gaussian noise, so that the mean and covariance of the cloud are preserved. Two safeguards depart from it: a variance floor that decays with the number of measurements, so early rejuvenations can still widen the cloud, and reflection of particles that leave the prior bounds. Each particle's noise posterior is then recomputed from the full history at its new position. The initial cloud is drawn from the prior. The formulas and settings are in Sec. S6 of the Supplemental Material [31].

The particle filter reports the posterior mean and the weighted particle standard deviation,

$$\hat{\theta}_N = \sum_i w_i \theta_i, \qquad \hat{\sigma}_N = \Big[\sum_i w_i (\theta_i - \hat{\theta}_N)^2\Big]^{1/2}, \quad (25)$$

the claimed uncertainty whose meaning Sec. III C discusses.

B. Measurement strategies

We compare two families of strategies. The sweep family holds the strategies an experimentalist would use without adaptive design: plain sweeps at several grid spacings, and Smart-Sweep, a sweep refined inside the dip once it is seen. The question is whether the best member of that family can match Bayesian SBED. All strategies infer with the particle filter of Sec. III A, so the comparison isolates the choice of measurements from the inference, and $\hat{\sigma}_N$ means the same for all of them.

All run under the same stop rule and the same measurement budget: a run stops once the claimed uncertainty of every parameter has stayed below its ceiling for several consecutive measurements and the dip test below finds a dip.

Dip test. The test asks where the measurements, not the particles, show a dip. Each measurement is scored against a baseline taken from the upper part of the data, in units of the noise from the noise posterior. Contiguous stretches of measured drive frequencies, about as wide as a plausible line, get a boxcar score, and a stretch is significant when its score passes a family-wise false-alarm threshold over all stretches tested. Each connected run of significant stretches is one dip.

Sweep family. SimpleSweep measures a uniform grid over the window, a few points per FWHM of the prior-mean line, and repeats whole passes; the spacing is the one setting a sweep has, and several are run. SmartSweep sweeps the window coarsely until the dip test finds a dip, then finely inside the focus (below), and then measures only at the steepest-slope points of the posterior-mean line, once the posterior expects them to carry nearly the optimal information. Its coarse search uses the same grid points, and the same measured values, as the reference sweep of that repeat.

Bayesian SBED. SBED is the particle filter choosing each d by Eq. (18), maximized over candidate drive frequencies concentrated on the steep sides of the posterior-mean dip and on the dips found. The candidates are restricted to the focus: one section per detected dip, padded by one half width on each side, which widens at once when a new dip appears and narrows only after a delay. A strategy that has piled its measurements onto a suspected dip cannot be contradicted elsewhere, so with a small probability, largest while much of the window is unmeasured, SBED instead measures the middle of the widest unmeasured gap, a falsification probe.

C. Simulation setup and evaluation

What the claimed uncertainty is. The particle filter reports $\hat{\sigma}_N$, Eq. (25): a moment of the posterior, defined at every N. Its run average is compared with the bound from the first measurement on (Sec. II D). What fails before the posterior is Gaussian (Sec. II F) is the usual reading of $\hat{\sigma}_N$: with a uniform window or several live candidates it is not a 68% interval, and the posterior entropy is below $\frac{1}{2}\ln(2\pi e\,\hat{\sigma}_N^2)$, so information gains, per-step gains as variance ratios, and coverage at Gaussian quantiles are only meaningful in the data regime. We therefore (i) compare $\hat{\sigma}_N$ with the bound as a run average at all N, (ii) test coverage, the true error against $\hat{\sigma}_N$, only once the posterior is Gaussian, or with quantiles of the particle distribution itself, and (iii) keep this separate from miscalibration of the particle filter, which makes $\hat{\sigma}_N$ too small at any N.

Use cases. A use case is one simulated signal at one noise level, set by the SNR, the drive window, the Gaussian share of the width and the isotope (Sec. S6 of the Supplemental Material [31]). Every metric is computed within one use case, over repeats whose true signal is the same for every strategy, so strategies are compared run by run; several use cases are summarized by the median over them.

Metrics. The accuracy at the end of a run is not a quality metric: every run stops once its claimed uncertainty is below the ceiling, so its final error mostly reflects the ceiling. We report instead whether the claimed uncertainty is honest, through its coverage, the share of runs whose true error lies within $\hat{\sigma}_N$ and within $2\hat{\sigma}_N$; how close each strategy comes to the bound of Eq. (13), through the ratio of the bound to the mean squared true error at each N; and the cost, the number of measurements after which the true error stays below a target. Each run is also read against the bound through its information rate, the growth of $\hat{\sigma}_N^{-2}$ per measurement in units of $I_{\max}$, Eq. (10); a rate above 1 means the claimed uncertainty fell faster than the data allow. Measurements to stop are compared only between strategies whose claimed uncertainty is equally honest.

D. Experimental setup

We measure the lower transition f_- only: in our setup it needs less microwave power than f_+, which keeps the heating of the stage low and the amplifier within its linear range. [Sample, optics, microwave delivery, detection and the per-shot integration time t_s.]

IV. RESULTS

A. Simulations

[(a) Claimed σ and true error against Eqs. (13)–(16) for a narrow and a wide window. (b) Information rate and prior share. (c) Cost vs SNR and window. (d) Calibration of the claimed σ.] [When the SBED posterior becomes Gaussian, and its calibration in refine (Sec. II F).] [Test: share of localize-stage measurements placed more than a few FWHM from the peak, compared with the residual mass there (Sec. II F).]

B. Experiment

[Placeholder until the lab data are in.]

V. DISCUSSION

[Limits: the claimed σ is overconfident; the simulated prior is miscalibrated; Gaussian noise only; $I_{\max}$ assumes

known nuisances; low-SNR detection model.]

VI. CONCLUSION

[Conclusion.]

Appendix A: Steepest slope of the Lorentzian line

This appendix derives the best information per measurement, Eq. (10), for the Lorentzian line of Eq. (4). Written in terms of the detuning $u = d - \theta$ of the measurement from the resonance, the line depends on d and θ only through u, so a shift of the resonance acts like the opposite shift of the measurement, and the slope of Eq. (9) is

$$\frac{\partial\mu}{\partial\theta} = -\frac{\partial\mu}{\partial u} = -\frac{2C\gamma^2 u}{(\gamma^2+u^2)^2}.$$

It vanishes at $u = 0$ and as $u \to \pm\infty$. Setting $\partial^2\mu/\partial u^2 = 0$ gives $(\gamma^2+u^2)^2 = 4u^2(\gamma^2+u^2)$, i.e. $u^2 = \gamma^2/3$: the slope is steepest at $|u| = \gamma/\sqrt{3}$, where $\gamma^2/(\gamma^2+u^2) = 3/4$ and the dip is 3/4 deep. There

$$\max_d \left|\frac{\partial\mu}{\partial\theta}\right| = \frac{2C\gamma^2\,(\gamma/\sqrt{3})}{(4\gamma^2/3)^2} = \frac{9C}{8\sqrt{3}\,\gamma} = \frac{9C}{4\sqrt{3}\,\Gamma} \approx 1.30\,\frac{C}{\Gamma},$$

with $\Gamma = 2\gamma$. Inserting this slope into Eq. (9), with $\mathrm{SNR} = C/\sigma$,

$$I_{\max} = \frac{k}{\sigma^2}\Big(\frac{9C}{4\sqrt{3}\,\Gamma}\Big)^2 = \frac{81}{48}\,\frac{k\,C^2}{\sigma^2\Gamma^2} = \frac{27}{16}\,k\,\frac{\mathrm{SNR}^2}{\Gamma^2}.$$

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